CO3-AT1

CASE STUDY

DSA0405

FUNDAMENTALS OF DATA SCIENCE

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Case Study 6 — Detecting an Unusual Customer

Problem Statement

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400. A particular customer spends ₹3,200. The company wants to determine whether this customer's spending behaviour is unusually high compared with other customers and understand what different z-score values indicate.

Questions

- 1. Calculate the customer's z-score.
- 2. Is the customer's spending unusually high compared with the average?
- 3. What would a negative z-score mean in this context?
- 4. If another customer has a z-score of −1.5, what does it tell you?

Step-by-Step Solution

Given Data:

- • **Mean (μ):** ₹2,000
- • **Standard Deviation (σ):** ₹400
- • **Customer Spending (x):** ₹3,200

Calculate the Z-score

Step 1: Write the Formula

$z = (x - \mu) / \sigma$

Step 2: Substitute the Given Values

$z = (3200 - 2000) / 400$

Step 3: Perform the Subtraction

$3200 - 2000 = 1200 \rightarrow z = 1200 / 400$

Step 4: Divide by the Standard Deviation

$1200 / 400 = 3$

> **Answer: Z-score = 3.** This means the customer's spending is 3 standard deviations above the average spending amount.

Is the Customer’s Spending Unusually High?

A z-score tells us how far a value is from the mean.

Z-score	Meaning
0	Exactly average
+1	One standard deviation above average
+2	Two standard deviations above average
+3	Three standard deviations above average
−1	One standard deviation below average

Since the customer has $z = 3$, the spending amount is far above the average.

Statistical Interpretation

In many statistical analyses:

- • **$|z| < 2$:** Normal variation
- • **$2 \leq |z| < 3$:** Unusually high or low
- • **$|z| \geq 3$:** Very unusual or extreme

Because $z = 3$, the customer belongs to the extreme high-spending category.

Conclusion: Yes, the customer’s spending is unusually high compared with the average customer spending.

What Does a Negative Z-score Mean?

A negative z-score means the customer spends less than the average amount.

Example 1: Suppose a customer spends ₹1,600.

$z = (1600 - 2000) / 400 = -400 / 400 = -1$

Interpretation: A z-score of −1 means the customer spends one standard deviation below the average.

Example 2: If a customer spends ₹1,200:

$z = (1200 - 2000) / 400 = -800 / 400 = -2$

This customer spends much less than the average customer.

Meaning in This Context — A negative z-score indicates:

- Lower spending than average
- Possibly an occasional customer
- Reduced purchasing activity
- A customer who may need promotional offers to increase engagement

Interpretation of $z = -1.5$

A z-score of -1.5 means the customer's spending is 1.5 standard deviations below the mean.

Step 1: Use the Formula $x = \mu + z\sigma \rightarrow x = 2000 + (-1.5 \times 400)$

Step 2: Multiply $\rightarrow -1.5 \times 400 = -600$

Step 3: Add to the Mean $\rightarrow x = 2000 - 600 =$ ₹1,400

Interpretation: A customer with $z = -1.5$ spends approximately ₹1,400. This means spending is below average, less than most customers, but not necessarily an extreme outlier (moderately low, not exceptionally low).

Detailed Explanation: Why Businesses Use Z-scores

E-commerce companies analyse customer spending patterns to make better business decisions. Z-scores help identify premium customers, bulk purchasers, frequent shoppers, inactive customers, suspicious transactions, and fraudulent behaviour.

Z-score Range	Customer Type
$z \geq 3$	VIP / Premium customer
$1 \leq z < 3$	Above-average customer
$-1 < z < 1$	Average customer
$-3 < z \leq -1$	Below-average customer
$z \leq -3$	Extremely low-spending customer

The customer with z = 3 may be purchasing expensive products, buying multiple items at once, making a festival/seasonal purchase, or is a loyal high-value customer.

Business Importance:

- • **For High Z-score Customers:** Offer loyalty rewards, exclusive discounts, personalized recommendations, invitation to premium memberships, and priority customer support.
- • **For Low Z-score Customers:** Send promotional coupons, cashback incentives, low-cost product recommendations, and encourage repeat purchases.

Summary Table — Case Study 6

Sub Question	Result
Z-score	3
Is spending unusually high?	Yes
Negative z-score means	Spending below average
$z = -1.5$ means	1.5 standard deviations below average
Approximate spending for $z = -1.5$	₹1,400

Final Conclusion: The customer who spent ₹3,200 has a z-score of 3, which indicates that the spending amount is significantly higher than the average customer spending of ₹2,000. Such a value is considered unusually high and may indicate a premium or high-value customer. A negative z-score represents spending below the average, while a z-score of −1.5 indicates a customer who spends moderately less than most customers, approximately ₹1,400. Therefore, z-scores are a useful statistical tool for comparing customer spending behaviour with the overall population, identifying unusual customers, and helping businesses make data-driven marketing and customer management decisions.

Case Study 7 — Delivery Time Analysis

Problem Statement

A delivery company records the following delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

The company wants to analyze the delivery performance and identify whether any delivery time is unusually high.

Questions

- 1. Calculate the mean and median.
- 2. Find Q1 and Q3.
- 3. Calculate the IQR.
- 4. Check for outliers using the 1.5 × IQR rule.
- 5. Explain whether the mean or median gives a better description of the typical delivery time.

Step-by-Step Solution

Given Data: 25, 27, 28, 29, 30, 30, 31, 32, 34, 60 | Number of observations (n) = 10

Calculate the Mean and Median

Step 1: Find the Mean

Sum = 25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326

Mean = 326 / 10 = 32.6 minutes

Step 2: Find the Median

Since n = 10 (even number), median is the average of 5th and 6th values (30 and 30).

Median = (30 + 30) / 2 = 30 minutes

Find Q1 and Q3

Step 1: Find Q1 (Lower half: 25, 27, 28, 29, 30) → Middle value Q1 = 28

Step 2: Find Q3 (Upper half: 30, 31, 32, 34, 60) → Middle value Q3 = 32

Calculate the IQR

Formula: IQR = Q3 − Q1 = 32 − 28 = 4

Check for Outliers

Step 1: Calculate 1.5 × IQR = 1.5 × 4 = 6

Step 2: Find Lower and Upper Limits:

- • **Lower Limit = Q1 − 6 = 28 − 6 =** 22
- • **Upper Limit = Q3 + 6 = 32 + 6 =** 38

Step 3: Compare Data Points:

Values 25 to 34 are within range [22, 38]. Value 60 > 38.

> **Outlier Identified:** 60 is an outlier because it is greater than the upper limit of 38.

Which Measure is Better?

Comparing Mean (32.6) and Median (30): The mean is higher because the value 60 pulls it upward. The median is less affected by extreme values.

Better Measure: Median is the better measure because it represents the typical delivery time more accurately when an outlier is present.

Detailed Explanation: Why the Outlier Matters

Most deliveries take between 25 and 34 minutes. The value 60 minutes is unusually high and may represent traffic congestion, vehicle breakdown, wrong address, or weather-related delays. Because of this unusual value, the mean becomes larger than the actual typical delivery time.

Summary Table — Case Study 7

Sub Question	Result
Mean	32.6 minutes
Median	30 minutes
Q1	28

Q3	32
IQR	4
Outlier	60
Better Measure	Median

Final Conclusion: The delivery data has a mean of 32.6 minutes and a median of 30 minutes. The quartiles are Q1 = 28 and Q3 = 32, giving an IQR of 4. Using the 1.5 × IQR rule, the value 60 minutes is identified as an outlier. Since the outlier increases the mean, the median provides a more accurate description of the typical delivery time for this dataset.

Case Study 8 — Two Cities' Temperatures

Problem Statement

The average daily temperatures (°C) for 7 days are:

- **City A:** 28, 29, 30, 30, 31, 30, 32
- **City B:** 20, 25, 30, 35, 30, 25, 45

Questions

- 1. Calculate the mean temperature for each city.
- 2. Calculate the variance for each city.
- 3. Which city has greater temperature variability?
- 4. Why can two datasets have the same mean but different standard deviations?
- 5. Which city has more predictable temperatures?

Step-by-Step Solution

Calculate the Mean Temperature

City A Mean: (28 + 29 + 30 + 30 + 31 + 30 + 32) / 7 = 210 / 7 = 30°C

City B Mean: (20 + 25 + 30 + 35 + 30 + 25 + 45) / 7 = 210 / 7 = 30°C

Calculate the Variance

Formula: Variance = $\Sigma(x - \bar{x})^2 / n$, where $\bar{x}$ = 30°C

Variance Calculations Table:

City A (x)	(x − 30)	(x − 30)²	City B (x)	(x − 30)²
28	-2	4	20	100
29	-1	1	25	25
30	0	0	30	0
30	0	0	35	25
31	1	1	30	0
30	0	0	25	25
32	2	4	45	225
Sum	-	10	Sum	400

City A Variance = 10 / 7 ≈ 1.43

City B Variance = 400 / 7 ≈ 57.14

Which City Has Greater Variability?

Comparing variances: City A = 1.43, City B = 57.14. Since 57.14 is much larger, City B has greater temperature variability.

Same Mean, Different Standard Deviations

Both cities have the same mean of 30°C. However, City A temperatures stay close to 30°C, whereas City B temperatures vary widely from 20°C to 45°C. This shows that the mean only measures the center, while the standard deviation/variance measures the spread of the data.

Which City Has More Predictable Temperatures?

A smaller variance means temperatures are more consistent. City A has more predictable temperatures because its values remain very close to 30°C throughout the week.

Summary Table — Case Study 8

Sub Question	City A	City B
Mean	30°C	30°C
Variance	1.43	57.14
Variability	Low	High
Predictability	More predictable	Less predictable

Final Conclusion: Although City A and City B have the same mean temperature of 30°C, their variability is very different. City A has a small variance (1.43), indicating stable and predictable weather, while City B has a large variance (57.14), indicating highly fluctuating temperatures. This demonstrates that datasets with the same mean can have very different levels of variability, so both the mean and variance must be considered when analyzing data.

Case Study 9 — Data Center CPU Usage

Problem Statement

A data scientist records the CPU usage percentages of a data center server during 10 monitoring intervals:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

The objective is to analyze the CPU usage pattern, identify any unusual values, and determine the overall distribution of the data.

Questions

- 1. Calculate the mean, median, and mode.
- 2. Calculate the range.
- 3. Find Q1, Q3, and IQR.
- 4. Identify possible outliers.
- 5. Determine the direction of skewness.

Step-by-Step Solution

Given Data: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90 | Observations (n) = 10

Calculate Mean, Median, and Mode

Step 1: Find the Mean

Sum = 40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513

Mean = 513 / 10 = 51.3%

Step 2: Find the Median

Average of 5th and 6th values (46 and 48) → Median = (46 + 48) / 2 = 47%

Step 3: Find the Mode

Value occurring most frequently: 45 appears twice. Mode = 45%

Calculate the Range

Range = Maximum value − Minimum value = 90 − 40 = 50%

Find Q1, Q3, and IQR

Step 1: Q1 (Lower half: 40, 42, 45, 45, 46) → Q1 = 45

Step 2: Q3 (Upper half: 48, 50, 52, 55, 90) → Q3 = 52

Step 3: IQR = Q3 − Q1 = 52 − 45 = 7

Identify Possible Outliers

Step 1: 1.5 × IQR = 1.5 × 7 = 10.5

Step 2: Limits:

- • **Lower Limit = Q1 − 10.5 = 45 − 10.5 =** 34.5%
- • **Upper Limit = Q3 + 10.5 = 52 + 10.5 =** 62.5%

Step 3: Compare values: All values except 90 lie within [34.5, 62.5]. Since 90 > 62.5, 90% is a possible outlier.

Determine the Direction of Skewness

Most values are between 40 and 55, while one value (90) is much larger. Since Mean (51.3%) > Median (47%), the distribution is positively skewed (right-skewed).

Detailed Explanation: Why Is 90 Important?

A CPU usage of 90% is unusually high compared with other observations. Possible reasons include a sudden increase in user requests, heavy background processing, software updates, malware scans, or system overload. This extreme value increases the mean and skews the distribution, while the median remains stable.

Summary Table — Case Study 9

Measure	Result
Mean	51.3%
Median	47%
Mode	45%
Range	50%
Q1	45
Q3	52
IQR	7
Outlier	90%
Skewness	Positive (Right)

Final Conclusion: The CPU usage data has a mean of 51.3%, a median of 47%, and a mode of 45%. The range is 50%, showing considerable spread in the data. The quartiles are Q1 = 45 and Q3 = 52, giving an IQR of 7. Using the 1.5 × IQR rule, the value 90% is identified as an outlier. Because the outlier is on the higher side, the distribution is positively skewed. Therefore, the server generally operates around 45–52% CPU usage, but the 90% observation indicates a potentially abnormal or high-load event that should be investigated further.

Case Study 10 — Comparing Two Datasets

Problem Statement

Two data analysts receive the following datasets representing the prediction errors of two different models:

- **Dataset A:** 10, 20, 30, 40, 50
- **Dataset B:** 28, 29, 30, 31, 32

The analysts want to compare the datasets and determine which model is more consistent.

Questions

- 1. Calculate the mean of both datasets.
- 2. Calculate the variance of both datasets.
- 3. Calculate the standard deviation of both datasets.
- 4. Which dataset has greater spread?
- 5. If both datasets represent model errors, which model appears more consistent?
- 6. Explain why mean alone is not enough to compare datasets.

Step-by-Step Solution

Calculate Means:

- • **Dataset A:** Mean = (10 + 20 + 30 + 40 + 50) / 5 = 150 / 5 = 30
- • **Dataset B:** Mean = (28 + 29 + 30 + 31 + 32) / 5 = 150 / 5 = 30

Calculate Variance & Standard Deviation

Formula: Variance = $\Sigma(x - \bar{x})^2 / n$

Dataset A (x)	(x − 30)	(x − 30)²	Dataset B (x)	(x − 30)²
10	-20	400	28	4
20	-10	100	29	1
30	0	0	30	0
40	10	100	31	1
50	20	400	32	4
Sum	-	1000	Sum	10

Dataset A Variance = 1000 / 5 = 200 | Standard Deviation = $\sqrt{200} \approx 14.14$

Dataset B Variance = 10 / 5 = 2 | Standard Deviation = $\sqrt{2} \approx 1.41$

Which Dataset Has Greater Spread?

Comparing standard deviations (14.14 vs 1.41), Dataset A has a significantly greater spread.

Which Model Appears More Consistent?

If the datasets represent model prediction errors, Dataset B represents the more consistent model because its errors remain tightly clustered around 30.

Why Is Mean Alone Not Enough?

Both datasets have the same mean (30), but their behaviour is vastly different. Dataset A values are spread widely (10 to 50), making predictions less reliable. Dataset B values are tightly clustered (28 to 32), making predictions reliable. The mean only shows the center of data; variance and standard deviation are necessary to measure spread, consistency, and reliability.

Summary Table — Case Study 10

Measure	Dataset A	Dataset B
Mean	30	30
Variance	200	2
Standard Deviation	14.14	1.41
Spread	High	Low
Consistency	Less consistent	More consistent

Final Conclusion: Both datasets have the same mean of 30, but they differ greatly in their variability. Dataset A has a variance of 200 and a standard deviation of approximately 14.14, indicating that the values are widely spread out. Dataset B has a variance of 2 and a standard deviation of approximately 1.41, showing that its values are tightly clustered around the mean. If these datasets represent model prediction errors, Dataset B corresponds to the more consistent and reliable model because its errors remain very close to the average.